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Heterogeneous Integration and the AI Computing Challenge

How TSMC Is Building an Ecosystem Around Advanced Packaging

At SEMICON West 2024, semiconductor executives repeatedly used the metaphor of a “wall” to describe the technical challenges confronting high-performance computing for artificial intelligence. In fact, the industry faces two major walls. The first is power consumption. As AI systems demand ever greater computing performance, the energy required to operate them continues to rise. The second is the “memory wall”: the growing disparity between improvements in processor performance and the ability of memory systems to supply processors with data fast enough to take advantage of that performance.

Over the past two decades, computing performance has increased by roughly 90,000 times, while memory bandwidth has grown by only about 30 times. Generative AI makes that imbalance increasingly difficult to ignore. Training and operating AI systems requires enormous amounts of text, images, video, code, translation data, and other information to move continuously between processors and memory. The emergence of high-bandwidth memory, or HBM, is one industry response to this problem. But HBM by itself illustrates a broader transformation taking place across semiconductor technology: improving one component is no longer sufficient.

When the Package Becomes the Chip

The semiconductor industry continues to push wafer fabrication toward increasingly advanced process nodes, including technologies below two nanometers. Memory manufacturers are developing and ramping HBM. Networking companies are pursuing technologies ranging from InfiniBand to optical interconnects. Yet advances within any single technical domain can no longer satisfy the extraordinary, system-wide performance demands created by generative AI.

This is why heterogeneous integration has become increasingly important. While moderating a SEMICON West session on heterogeneous integration and packaging, William Chen of ASE Group summarized the shift with a concise observation: “The package is the chip.” The statement captures an important change in how advanced computing devices are conceived. A new-generation “chip” is no longer necessarily a single monolithic piece of silicon. Instead, multiple components manufactured for different functions can be integrated within one advanced package, creating a system whose performance depends as much on packaging and interconnection as on the individual dies themselves.

NVIDIA’s H100 provides an example. The product combines a GPU manufactured using TSMC’s 4-nanometer-class process with HBM memory inside a high-performance computing architecture. Newer generations push integration even further, effectively bringing multiple major computing and memory components together within increasingly sophisticated packages. Technologies such as TSMC’s CoWoS and Samsung’s advanced packaging platforms use 2.5D and 3D architectures to integrate components that previously would have been treated as separate devices.

The implication is significant. Packaging is no longer merely the final step after a chip has been manufactured. It is becoming part of the architecture of the computing system itself.

Chiplets Move Into the Automotive Industry

This heterogeneous-integration trend is not limited to AI data centers. It is also moving rapidly into automotive electronics—an industry traditionally known for taking an exceptionally cautious approach to new semiconductor technologies because reliability and safety requirements are so demanding.

At SEMICON West, the Smart Mobility program focused heavily on chiplets. The automotive industry is increasingly moving away from architectures in which functions such as engine control, reversing systems, cruise control, communications, and entertainment are handled by numerous independent processing systems. Instead, different functional chiplets can be integrated into more centralized computing platforms supporting vehicle control, communications, infotainment, electrification, and eventually more advanced autonomous-driving capabilities.

For an industry accustomed to carefully validated, incremental technological change, this represents a substantial architectural transition. Bosch's Crylle Brandon captured the industry's cautious mood when discussing the shift: "We are all very nervous." Nevertheless, pressures from electrification, digitalization, connectivity, and autonomous driving are making greater semiconductor integration increasingly difficult to avoid.

AI Becomes Both the Challenge and the Tool

The semiconductor industry is not only building hardware to support AI. It is also using AI to accelerate the development of that hardware. Automotive semiconductor companies at SEMICON West discussed using AI to improve chiplet development. Semiconductor foundries such as TSMC have been recruiting specialists in artificial intelligence and machine learning to improve manufacturing operations and maintain quality.

Further upstream, electronic design automation companies have demonstrated that AI can dramatically reduce some verification workloads—from processes measured in weeks to processes that can potentially be completed in hours. Semiconductor materials companies are also applying AI to accelerate the search for new materials that may enable more efficient generations of semiconductor devices.

This creates an unusual feedback loop. AI is placing unprecedented demands on processors, memory, packaging, networking, power systems, and semiconductor manufacturing. At the same time, the semiconductor industry is using AI itself to accelerate the design, verification, manufacturing, and materials research needed to overcome those challenges. Capturing the opportunity created by AI therefore requires participation across the entire semiconductor value chain.

TSMC Builds an Ecosystem Around 3DFabric

This need for coordination helps explain TSMC's approach to heterogeneous integration. To support the development and production of its CoWoS and other advanced-packaging technologies, TSMC has organized the 3DFabric Alliance, bringing together companies across multiple parts of the semiconductor ecosystem.

At SEMICON West, TSMC Senior Director Lluís Paris presented a view of this ecosystem to an audience of industry professionals. The participating fields span EDA design services, intellectual property, memory, substrates, outsourced assembly and testing, and other technologies required to make advanced heterogeneous systems work reliably.

The objective is not simply to assemble a list of suppliers. Advanced packaging requires participants across the ecosystem to coordinate technologies, interfaces, development schedules, manufacturing requirements, and product roadmaps. That coordination is increasingly important because heterogeneous integration changes where system-level engineering takes place.

In a conventional model, improvements could often be pursued within relatively distinct areas: a smaller process node for the processor, faster memory, or a better interconnect. In an AI system built around advanced packaging, however, the performance of the overall product increasingly depends on how those technologies interact. The package, memory, processor, interconnect, substrate, design tools, testing processes, and manufacturing capabilities become parts of the same engineering problem.

Beyond Moore's Law: Integration as a System-Level Strategy

The surge in AI investment following the emergence of ChatGPT in late 2022 has made the industry's technical constraints much more visible. Increasing transistor density remains important, but the semiconductor industry's response to AI cannot depend on process scaling alone. The memory wall, power consumption, data movement, packaging complexity, and system integration all have to be addressed simultaneously.

Heterogeneous integration offers one way forward by allowing different technologies to be combined according to their strengths rather than requiring every function to be built into a single monolithic device. But this approach also changes the nature of semiconductor competition. Success increasingly depends not only on who can manufacture the most advanced individual component, but also on who can coordinate an ecosystem capable of integrating processors, memory, chiplets, substrates, packaging, interconnects, design tools, and testing into reliable systems at scale.

TSMC's decision to build the 3DFabric Alliance reflects this transition. The company is positioning advanced packaging not simply as another manufacturing service, but as an ecosystem in which multiple semiconductor technologies must evolve together. AI may provide the demand that drives the semiconductor industry toward its next major growth milestone, but it is simultaneously raising the technological barriers the industry must overcome.

Breaking through the power wall and the memory wall will require advances across the entire value chain. In that environment, heterogeneous integration is no longer simply a packaging technology. It is becoming one of the fundamental architectural strategies for the next generation of computing.

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